

Django Fundamentals and the MTV Architecture

What Is Django?

Django is a high-level Python web framework used to build secure, scalable, and maintainable web applications. It provides built-in features such as authentication, database management, and an admin interface.

- Built with Python
- Fast development
- Strong security
- Works with databases
- Clean structure

MTV Architecture

Django uses the Model-Template-View (MTV) architecture to separate concerns in web applications.

Model

Defines data structure and database interaction.

- Represents database tables
- Stores and retrieves data
- Defines fields and relationships

Template

Handles presentation and user interface.

- Displays dynamic content
- Uses HTML with Django syntax
- Separates design from logic

View

Handles logic and connects models with templates.

- Processes requests
- Retrieves data from models

- Returns responses

MTV Flow

- User sends request
- URL maps to view
- View interacts with model
- Data passed to template
- Template renders response

Core Django Commands

- `pip install django`
- `django-admin startproject myproject`
- `python manage.py runserver`

Conclusion

Django simplifies web development through its structured MTV architecture, making applications easier to build, maintain, and scale.